

Deep Generative Models

Lecture 12

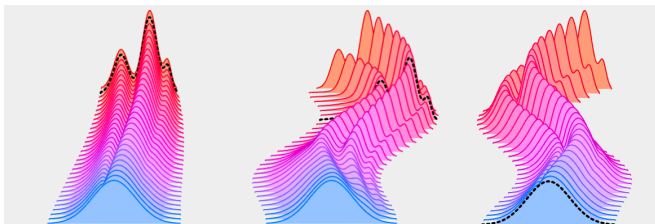
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Recap of Previous Lecture

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$



Problem: The true vector field $\mathbf{v}(\mathbf{x}, t)$ is **unknown**.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})).$$

Recap of Previous Lecture

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimum for CFM.

Training

1. Sample $t \sim U[0, 1]$, $\mathbf{z} \sim p(\mathbf{z})$, $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L} = \|\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Recap of Previous Lectures

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})} \mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

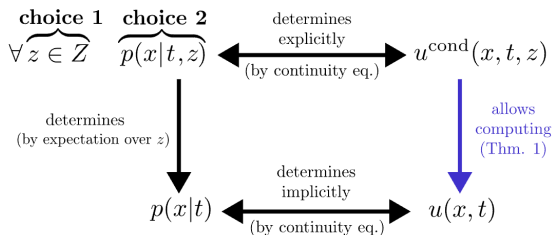
Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimum for CFM.

Recap of Previous Lecture



Constraints

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Q1 How should we choose the convenient conditioning latent variable $\mathbf{z}$?

Q2 How can we parametrize $p_t(\mathbf{x}|\mathbf{z})$ to enforce the constraints?

Gaussian Probability Path and Flow

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})) \quad \mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

image credit: *A Visual Dive into Conditional Flow Matching*

Recap of Previous Lecture

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})); \quad \mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\psi}'_t(\boldsymbol{\psi}_t(\mathbf{x}_t, \mathbf{z})^{-1}, \mathbf{z}) = \boldsymbol{\mu}'_t(\mathbf{z}) + \frac{\boldsymbol{\sigma}'_t(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

Conditioning Latent Variable

Let's choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_1)p_1(\mathbf{x}_1)d\mathbf{x}_1$$

We must ensure the boundary constraints:

$$\begin{cases} \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}) \\ \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}) \end{cases} \Rightarrow \begin{cases} p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \\ p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1). \end{cases}$$

Recap of Previous Lecture

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1).$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \mathbf{x}_0.$$

Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1; \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t. \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I}); \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0. \end{cases}$$

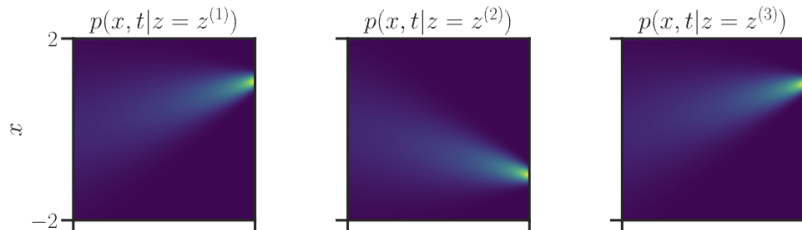


image credit: A Visual Dive into Conditional Flow Matching

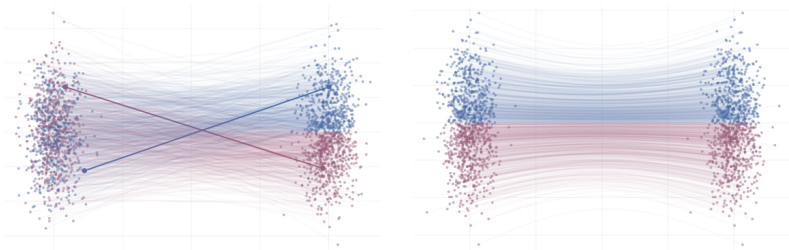
Recap of Previous Lecture

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t} = \mathbf{x}_1 - \mathbf{x}_0$$

$$\begin{aligned} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 &= \\ = \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, t)\|^2 \end{aligned}$$

- ▶ $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.
- ▶ The **marginal** path $p_t(\mathbf{x})$ does not give straight lines.



Recap of Previous Lecture

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2 \rightarrow \min_{\theta}$$

Training

1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
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$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Outline

1. Two-Sided Conditioning
2. Link between Flow Matching and Score-Based Models
3. Discrete Diffusion Models
4. Forward Discrete Process
5. Reverse Diffusion Process
6. From Token To Sequence

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Pair Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$. Then $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1)p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)d\mathbf{x}_0d\mathbf{x}_1$$

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We must enforce boundary constraints:

$$\begin{cases} p_0(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \\ p_1(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) \end{cases}$$

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Two-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

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Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \\ \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) = \epsilon \end{cases}$$

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Two-Sided Conditioning

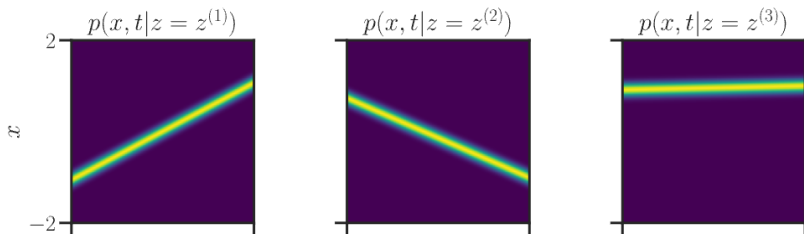
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$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

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Flow Matching: One-Sided vs. Two-Sided Conditioning

$$\mathbf{z} = \mathbf{x}_1$$

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

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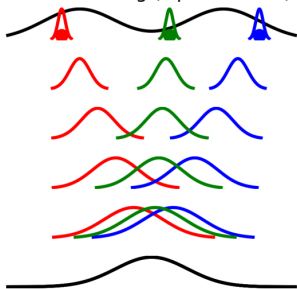
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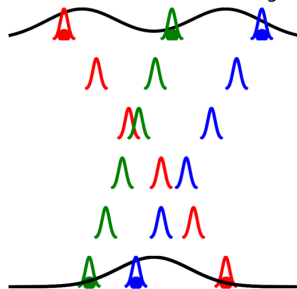
$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I})$$

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Flow Matching (Lipman et al.)



Conditional Flow Matching



Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_0, \mathbf{x}_1)}{\sigma_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_0, \mathbf{x}_1))$$

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\begin{aligned} \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) &= \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1)) \\ \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) &= \mathbf{x}_1 - \mathbf{x}_0 \end{aligned}$$

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N} \left(t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, \epsilon^2 \mathbf{I} \right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$$

Conditional Vector Field

$$\begin{aligned} \frac{d\mathbf{x}_t}{dt} &= \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1)) \\ \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) &= \mathbf{x}_1 - \mathbf{x}_0 \end{aligned}$$

Conditional Flow Matching

$$\begin{aligned} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1)} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \end{aligned}$$

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

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$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$$

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 =$$
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1)} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2$$

- ▶ This yields the same procedure as for one-sided conditioning!
- ▶ Now, we do not require that $p_0(\mathbf{x})$ is necessarily $\mathcal{N}(0, \mathbf{I})$.

CFM with Two-Sided Conditioning: Training and Sampling

- ▶ This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- ▶ It's possible to apply this approach to paired tasks, e.g., style transfer.

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Training

1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

CFM with Two-Sided Conditioning: Training and Sampling

- ▶ This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- ▶ It's possible to apply this approach to paired tasks, e.g., style transfer.

Training

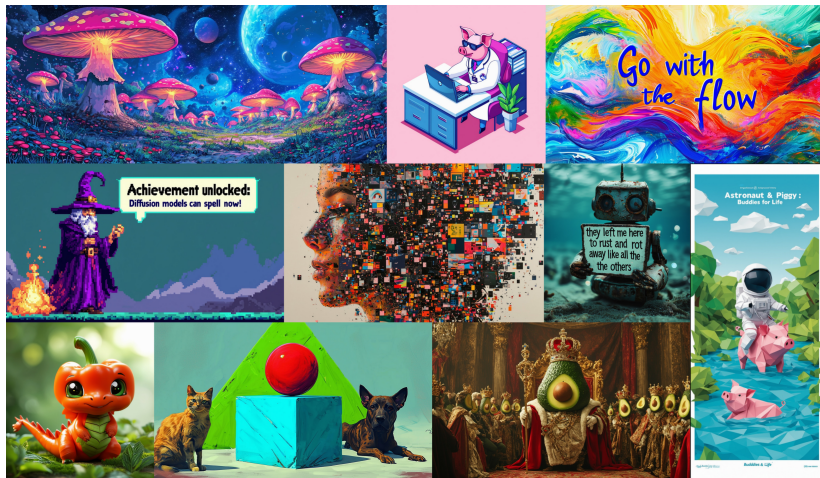
1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim p_0(\mathbf{x})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Stable Diffusion 3: Scalable Flow Matching



Esser P., et al. *Scaling Rectified Flow Transformers for High-Resolution Image Synthesis*, 2024

Outline

1. Two-Sided Conditioning
2. Link between Flow Matching and Score-Based Models
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Score-Based Generative Models through SDEs

Training

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

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Variance Exploding SDE (NCSN)

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)] \cdot \mathbf{I}), \quad \sigma(0) = 0$$

Variance Preserving SDE (DDPM)

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(0)\alpha(t), (1 - \alpha(t)^2) \cdot \mathbf{I}); \quad \alpha(t) = e^{-\frac{1}{2} \int_0^t \beta(s) ds}$$

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Flow matching uses reverse time direction:

$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Score-Based Generative Models through SDEs

$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

$$\mathbf{VE} \text{ (NCSN): } p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_1, \sigma_{1-t}^2 \cdot \mathbf{I})$$

$$\mathbf{VP} \text{ (DDPM): } p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2) \cdot \mathbf{I})$$

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Flow Matching Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I}); \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t}$$

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_1)}{\sigma_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_1))$$

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Let's derive the conditional vector fields for VE (NCSN) and VP (DDPM).

Flow Matching vs Score-Based SDE Models

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Variance Exploding SDE Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_1, \sigma_{1-t}^2 \mathbf{I}) \quad \Rightarrow \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

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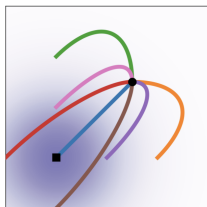
Variance Preserving SDE Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2)\mathbf{I}) \Rightarrow \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1 - \alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$

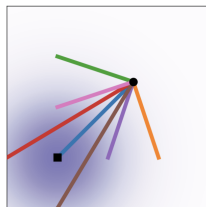
Thus, VE/VP SDE models correspond to particular choices of the Gaussian probability path within the flow matching framework.

Flow Matching vs Score-Based SDE Models

Trajectories



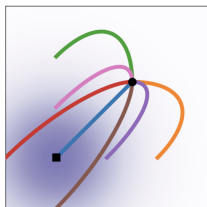
Diffusion



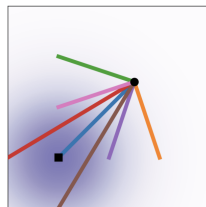
OT

Flow Matching vs Score-Based SDE Models

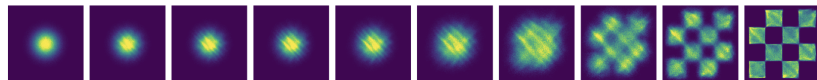
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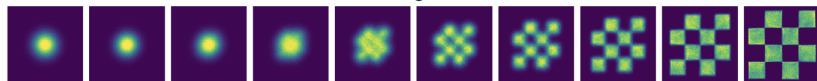
Diffusion



OT



Score matching ^{w/} Diffusion

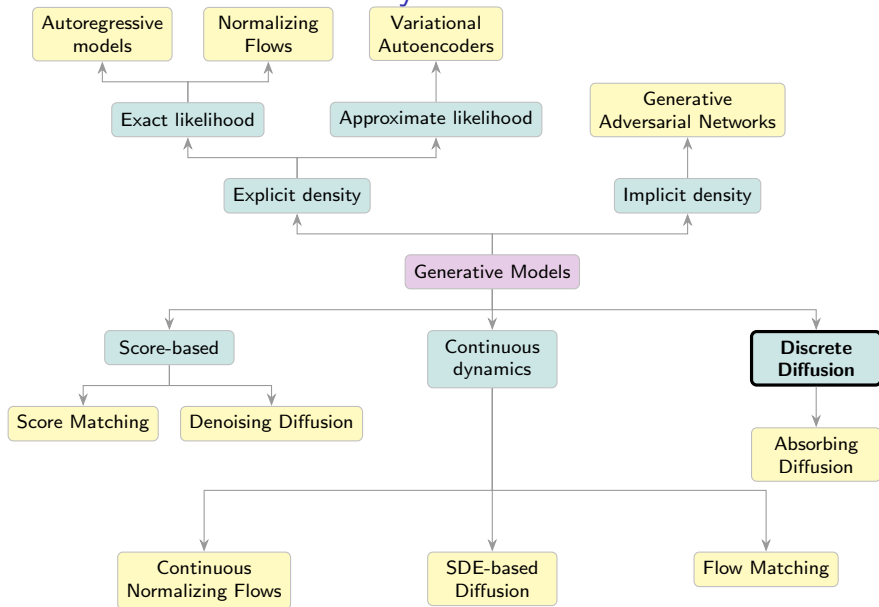


Flow Matching ^{w/} OT

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Generative Models Taxonomy



Discrete or Continuous Diffusion Models?

Reminder: Diffusion models define a forward corruption process and a reverse denoising process. Previously, we studied diffusion models with continuous states $\mathbf{x}(t) \in \mathbb{R}^m$.

Continuous state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\} \Rightarrow$ **DDPM / NCSN**.
- ▶ **Continuous time** $t \in [0, 1] \Rightarrow$ **Score-based SDE models**.

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Now we turn to diffusion over discrete-value states $\mathbf{x}(t) \in \{1, \dots, K\}^m$.

Discrete state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\}$.
- ▶ **Continuous time** $t \in [0, 1]$.

Let's discuss why we need discrete diffusion models.

Why Discrete Diffusion Models?

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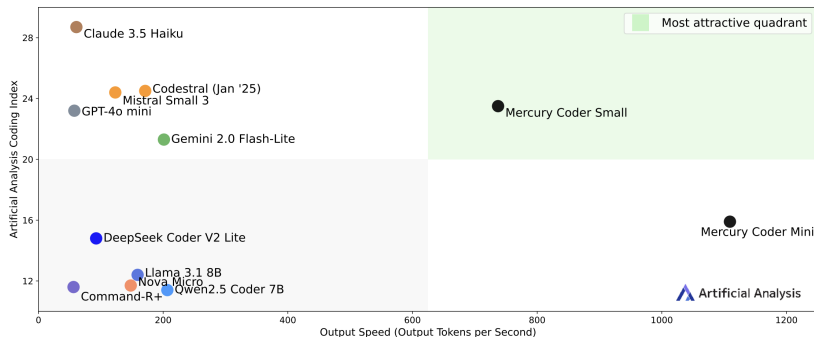
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- ▶ **Unified framework:** diffusion generalizes naturally to discrete domains that do not suit continuous Gaussian noise.

2025 – Big Bang of Discrete Diffusion Models

Coding Index vs. Output Speed: Smaller models

Artificial Analysis Coding Index (represents the average of LiveCodeBench & SciCode);
Output Speed: Output Tokens per Second; 1,000 Input Tokens; Coding focused workload



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Forward Discrete Process

Continuous Diffusion Markov Chain

In continuous diffusion, the forward Markov chain is defined by progressively corrupting data with Gaussian noise:

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For discrete data, we instead define a Markov chain over categorical states:

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- ▶ Each $\mathbf{x}_t \in \{0, 1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).
- ▶ What is the transition matrix $\mathbf{Q}_t$?

Forward Process over Time

Transition Matrix

$\mathbf{Q}_t \in [0, 1]^{K \times K}$ is a **transition matrix** where each column gives transition probabilities from one state to all others, and columns sum to 1:

$$[\mathbf{Q}_t]_{ij} = q(x_t = i | x_{t-1} = j), \quad \sum_{i=1}^K [\mathbf{Q}_t]_{ij} = 1.$$

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- ▶ As $t \rightarrow T$, the process drives the data toward a stationary distribution.
- ▶ We design the transition matrices $\mathbf{Q}_t$ to achieve this behavior.

Transition Matrix

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Common choices

- ▶ **Uniform diffusion**

$$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}, \quad \mathbf{U}_{ij} = \frac{1}{K}.$$

Each token is replaced by a uniformly random symbol with probability β_t . The stationary distribution is uniform noise.

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- ▶ **Absorbing diffusion**

$$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top.$$

Tokens are gradually replaced by a special mask m ; the stationary distribution is fully masked.

Transition Matrix

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{x}_0), \quad \mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

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- ▶ Each token retains its original value with prob. $\bar{\alpha}_t$.
- ▶ It becomes uniformly random with prob. $(1 - \bar{\alpha}_t)$.
- ▶ As $t \rightarrow T$, the process converges to the stationary uniform distribution.

Transition Matrix

Absorbing Diffusion

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- ▶ Each token retains its original value with prob. $\bar{\alpha}_t$.
- ▶ It becomes $\mathbf{e}_m$ with prob. $(1 - \bar{\alpha}_t)$.
- ▶ As $t \rightarrow T$, all tokens converge to the mask state:
 $q(\mathbf{x}_T) \approx \text{Cat}(\mathbf{e}_m)$.
- ▶ This makes the process analogous to **masked language modeling**.

Uniform vs. Absorbing Transition Matrix

Aspect	Uniform Diffusion	Absorbing Diffusion
$\mathbf{Q}_t$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$	$\mathbf{U}$	$\text{Cat}(\mathbf{e}_m)$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

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Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

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Posterior of the Forward Process

ELBO

$$\begin{aligned}\mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

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- ▶ Conditioned reverse distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ played crucial role in the continuous-state diffusion model.
- ▶ It shows the probability of a previous state given the noisy state $\mathbf{x}_t$ and the original clean data $\mathbf{x}_0$.

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Discrete conditioned reverse distribution

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &= \frac{\text{Cat}(\mathbf{Q}_t) \cdot \text{Cat}(\mathbf{Q}_{1:t-1})}{\text{Cat}(\mathbf{Q}_{1:t})}. \end{aligned}$$

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Recall the ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}\left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\right),$$

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- ▶ Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- ▶ The reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_\theta(\mathbf{x}_t, t)),$$

where $\boldsymbol{\pi}_\theta$ is a neural network.

Discrete-time ELBO for Discrete Diffusion

ELBO term

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Categorical KL

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \parallel \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}),$$

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- ▶ $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
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Therefore, minimizing $\mathcal{L}_t$ w.r.t. θ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Outline

1. Two-Sided Conditioning
2. Link between Flow Matching and Score-Based Models
3. Discrete Diffusion Models
4. Forward Discrete Process
5. Reverse Diffusion Process
6. From Token To Sequence

From Token to Sequence

One-hot sequence representation

$$\mathbf{x}_t \in \{0, 1\}^K \quad \Leftrightarrow \quad \mathbf{X}_t \in \{0, 1\}^{K \times m}$$

Here $\mathbf{X}_t$ is a one-hot representation of a sequence of tokens.

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Independent Token-wise Forward Process

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

- ▶ Each position i evolves according to its own Markov chain.
- ▶ Often the same transition matrix $\mathbf{Q}_t$ is shared across i .

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Continuous Diffusion Analogy

- ▶ In Gaussian DDPMs with diagonal covariance, noise is independent per pixel.
- ▶ Structure is not in the noise; it is learned by the reverse model.

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- ▶ All distributions defined by the forward process are factorized.
- ▶ Dependence appears in the learned reverse model.

Reverse Model for Sequence

$$p_{\theta}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- ▶ The output factorizes (parallel prediction across positions).
- ▶ Each factor conditions on the entire noisy sequence $\mathbf{X}_t$.
- ▶ This is exactly the **masked language modeling** pattern.

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Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^m \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[-\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right].$$

Summary

- ▶ Two-sided conditioning uses pair conditioning $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ and yields the same procedure, but is more general (suitable for paired tasks).
- ▶ Diffusion and score-based models are special cases of the flow matching approach, but use curved trajectories.
- ▶ Diffusion approach has several key advantages over autoregressive approach.
- ▶ Forward discrete diffusion process defines a Markov chain with discrete states; uniform and absorbing transitions make it tractable.
- ▶ Reverse discrete diffusion process uses the variational approach to invert forward process; the discrete-state ELBO is a cross-entropy loss.
- ▶ For sequences, the forward process factorizes over tokens, while the learned reverse model predicts all positions in parallel conditioning on the whole noisy sequence (masked LM pattern).